NESTED LOOP CALENDAR GENERATOR

Python Programming — Activity-Based Learning (ABL) Assignment

Student Name:	_____
Roll Number:	_____
Branch / Section:	_____
Course:	Python Programming

1. Aim

To develop a Python program that generates a neatly aligned monthly calendar using the number of days in a month and the weekday on which the month starts. Nested loops and conditional statements are used to format the dates into weeks.

2. Problem Statement

Create a formatted monthly calendar based on user-supplied starting weekday and number of days. The program must align dates under Sunday through Saturday, handle month lengths from 28 to 31 days, and display an error message for invalid inputs.

3. Objectives

• Accept the number of days and starting weekday from the user.

• Validate the day count (28–31) and weekday index (0–6).

• Align the first date beneath the correct weekday.

• Use loops and a conditional statement to print the calendar week by week.

4. Assumptions

Weekday numbering is Sunday = 0, Monday = 1, Tuesday = 2, Wednesday = 3, Thursday = 4, Friday = 5 and Saturday = 6. The input day count is between 28 and 31. This simple generator does not determine the month or year automatically.

5. Algorithm

1. Start the program.

2. Read the number of days and starting weekday.

3. If days are outside 28–31, print an invalid-day message. Otherwise, if weekday is outside 0–6, print an invalid-weekday message.

4. Print the weekday headings.

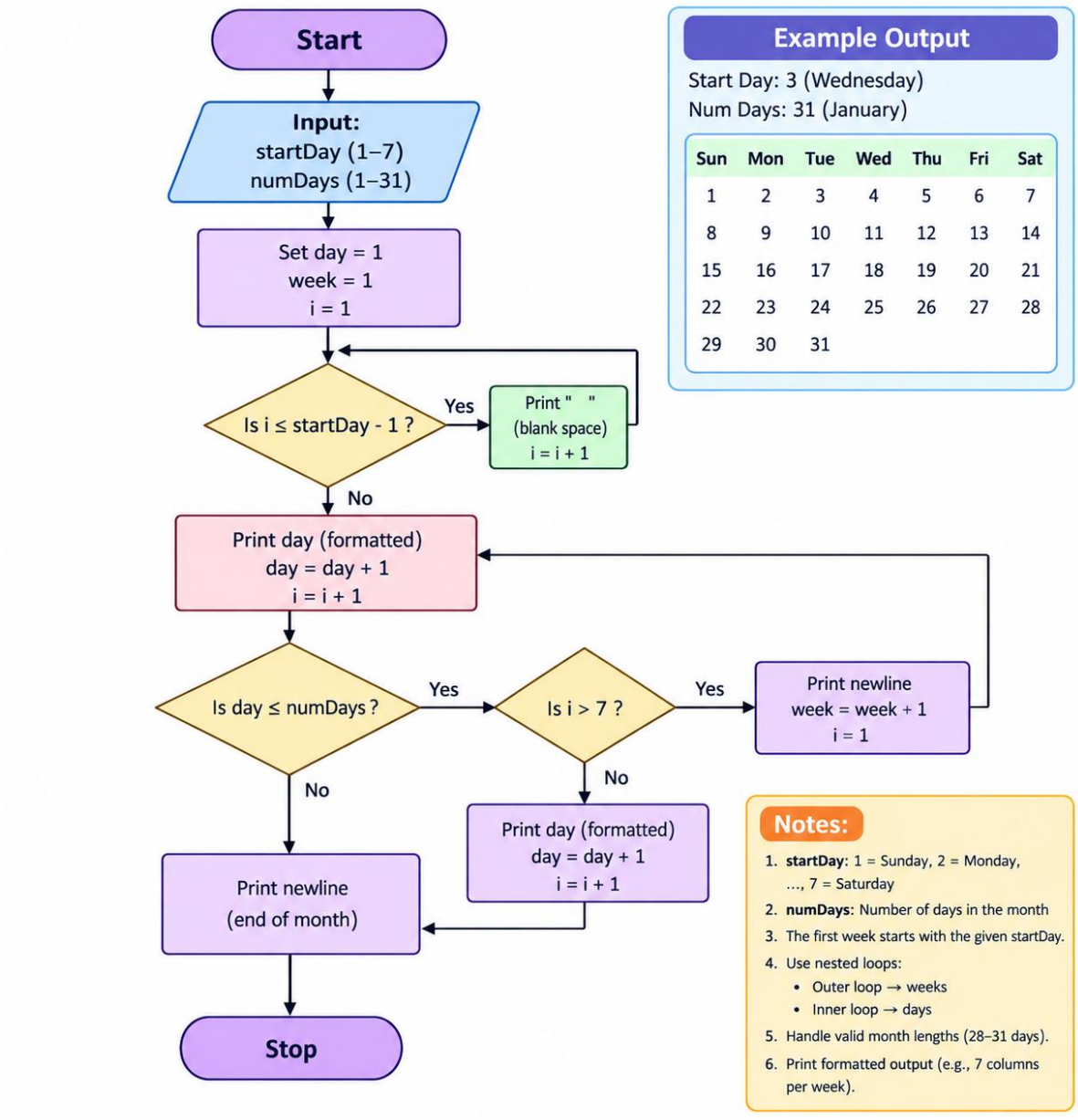
5. Print blank spaces for weekdays before day 1.

6. Loop from day 1 through the final day; print each day in a fixed-width field.

7. After every Saturday, move to the next line.

8. Print a final newline if the last week is incomplete, then stop.

6. Flowchart



7. pseudocode

```
BEGIN

  INPUT startDay
  INPUT numDays

  IF startDay < 1 OR startDay > 7 THEN
    PRINT "Invalid Starting Weekday"
    STOP
  END IF

  IF numDays < 28 OR numDays > 31 THEN
    PRINT "Invalid Number of Days"
    STOP
  END IF

  PRINT "Sun Mon Tue Wed Thu Fri Sat"

  day ← 1

  FOR week ← 1 TO 6

    FOR weekday ← 1 TO 7

      IF week = 1 AND weekday < startDay THEN
        PRINT "    "
```

```
        ELSE IF day <= numDays THEN
            PRINT day
            day ← day + 1
        ELSE
            BREAK
        END IF

    END FOR

    PRINT NEWLINE

    IF day > numDays THEN
        BREAK
    END IF

  END FOR

END
```

8. Python Source Code

```
# Nested Loop Calendar Generator

start_day = int(input("Enter starting weekday (1=Sun, 7=Sat): "))
num_days = int(input("Enter number of days in the month (28-31): "))

# Validate input
```

```
if start_day < 1 or start_day > 7:
    print("Invalid Starting Weekday")

elif num_days < 28 or num_days > 31:
    print("Invalid Number of Days")

else:
    # Print weekday headings
    print("\n Sun Mon Tue Wed Thu Fri Sat")

    day = 1

    # Outer loop: weeks
    for week in range(6):

        # Inner loop: weekdays
        for weekday in range(1, 8):

            if week == 0 and weekday < start_day:
                print("    ", end="")

            elif day <= num_days:
                print(f"{day:4}", end="")
                day += 1

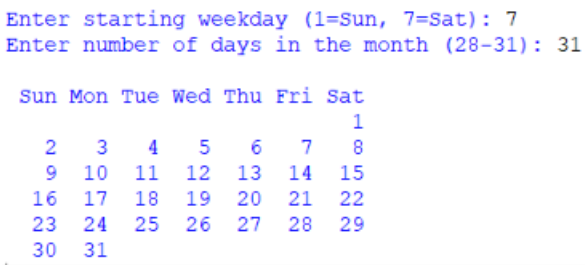
            else:
```

```
        break


    print()


    # Stop after printing all days
    if day > num_days:
       Break
```

9. Output Screenshot



10. Sample Test Case

1. **TC01 – Valid Input:** Verify that the calendar is generated correctly when the starting weekday and number of days are valid.

2. **TC02 – Sunday Start:** Check that the first date is aligned under Sunday when `start_day= 1`.

3. **TC03 – Monday Start:** Verify that the first date appears under Monday when `start_day = 2`.

4. **TC04 – Saturday Start:** Check that the first date is a under Saturday when `start_day = 7`.

5. **TC05 – 28-Day Month:** Verify that the calendar displays exactly 28 days.

6. **TC06 – 29-Day Month:** Check that the calendar correctly displays 29 days, including a leap-year February scenario.

7. **TC07 – 30-Day Month:** Verify that all 30 days are printed in the correct weekday columns.

8. **TC08 – 31-Day Month:** Check that all 31 days are displayed with proper weekly alignment.

9. **TC09 – Invalid Starting Weekday:** Verify that an error message is displayed when the starting weekday is less than 1 or greater than 7.

10. **TC10 – Invalid Number of Days:** Check that an error message appears when the number of days is less than 28 or greater than 31.

11. **TC11 – First-Week Alignment:** Verify that blank spaces are printed before the first date according to the supplied starting weekday.

12. **TC12 – Week Formatting:** Check that each calendar week is printed on a separate line with seven weekday columns.

13. **TC13 – Sequential Dates:** Verify that dates are printed in ascending order without missing or duplicated days.

14. **TC14 – End-of-Month Handling:** Check that the calendar stops immediately after the final date and does not print extra dates.

15. **TC15 – Boundary Values:** Verify that `start_day = 1`, `start_day = 7`, `num_days = 28`, and `num_days = 31` are handled correctly.

11. Conclusion

The Nested Loop Calendar Generator demonstrates how loops, conditional statements and fixed-width formatting can be combined to produce a readable monthly calendar. Input validation helps prevent unsupported day counts and weekday values.